Curriculum Vitae



Aatiq Ali

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Nationality: Indian | Date of Birth: 06.10.2001

WORK EXPERIENCE

Student Research Assistant & Master's Thesis Student Feb 2026- present
German Aerospace Center (DLR), Stuttgart, Germany

- Conducting master's thesis on **infrastructure-based sensor placement for autonomous driving**.
- Developing a Python-based, two-dimensional geospatial road-network placement approach for infrastructure sensors to support autonomous driving.
- Implementing spatial-processing and sensor-placement algorithms using Python, GeoPandas, Shapely, Pandas, and QGIS.
- Evaluating the reuse of streetlights, and traffic signals infrastructure across multiple urban study areas in Germany.
- Conducting parameter studies to analyse how varying coverage demands affect infrastructure reuse, and deployment cost.

Student Developer – Web Application Development (Academic Project) Jan 2022- Jul 2022
Department of Computer Science, VKCET College, Thiruvananthapuram, India

- Developed a **web-based Attendance Management System** with separate **login portals for parents, professors, and students**.
- Enabled professors to **mark and update attendance**, while students and parents could **view attendance records** through their own dashboards.
- Integrated an option to **download attendance data in Excel format** for reports and record-keeping.
- Focused on **HTML, CSS, and SQL** for user interface design and database management, ensuring smooth interaction and data security.

Intern – Mobile Application Development (Flutter) Jul 2021- Aug 2021
Fablab, Kochi, India

- Completed a practical internship focused on **Flutter-based mobile application development**.
- Designed simple **user interfaces** and created interactive pages for a prototype project.
- Learned about **cross-platform app development**, **UI layout**, and **basic backend connectivity** under mentor guidance.
- Worked collaboratively in a small team, gaining practical experience in **mobile app design** and project workflow.

EDUCATION

Master of Engineering (M.Eng.) in Autonomous Driving | *Coburg University of Applied Science, Coburg, Germany*.
March 2024 – Present **Grade:** 1.8

- Project-based curriculum with strong practical exposure to **Python programming, ROS, Deep Learning, Computer Vision, Sensor Fusion and Linux**.
- Hands-on experience with **LiDAR, camera, and NVIDIA Jetson** for perception and control systems.
- Key Project**:** *Group Start Platooning for Commuters* – developed **state-machine control logic**, worked with **V2X (SPaT) messages**, and applied **machine-learning models** for autonomous platoon coordination.

Bachelor of Technology in Computer Science and Engineering | *APJ Abdul Kalam University, Kerala, India*.
July 2019 – June 2023 **Grade:** 1.9

- Comprehensive academic foundation with theoretical coursework and practical laboratories in **AI, ML, DBMS, Microprocessors, Computer Graphics, Computer Networks and Data Structures**.
- **Final Year Project:** *Medical Record System using Blockchain Technology* – designed and implemented a secure platform for managing and sharing medical records, ensuring transparency and data integrity.

TECHNICAL SKILLS

Programming: Python, SQL, C, HTML, CSS.
Autonomous Systems: ROS 2, LiDAR, V2X/SPaT, sensor data processing, NVIDIA Jetson.
Geospatial Development: GeoPandas, Shapely, QGIS, OpenStreetMap
***Computer Vision & Machine Learning*:** OpenCV, Linear Regression, Decision Trees, SVM.
Development Tools & Platform: Git, GitHub, GitLab, Linux
***Design Tools*:** Figma, Canva, Adobe Photoshop.
Collaboration: Miro, Agile Methodology

PROJECTS

Robot State Machine

- Implemented a finite state machine in ROS 2 for navigation and task transitions in robotic systems, mimicking real vehicle behavior handling.
- Integrated live sensor data for responsive task switching, enhancing the understanding of real-time data processing similar to trace handling.
- Applied modular software design to improve code reusability, maintainability, and testing.

Obstacle Detection using LiDAR

- Built a ROS 2-based obstacle detection system using LiDAR sensor data.
- Processed real-time laser scan inputs to detect obstacles and publish alerts for safe navigation.
- Enhanced understanding of **sensor data filtering**, **distance thresholding**, and **real-time message publishing**.

Traffic Light Responder (V2X Communication)

- Developed a **ROS 2 node** that enables autonomous vehicles to respond intelligently to **traffic-light signals received through V2X communication**.
- Processed **SPaT (Signal Phase and Timing)** messages to extract real-time **event states** for red and green phases.
- Implemented logical decision routines to generate corresponding vehicle control responses for intersection handling.
- Enhanced understanding of V2X protocols, message parsing, and real-time decision control in cooperative traffic environments.

Medical Record System using Blockchain Technology

- Developed a **secure medical record management system** combining **centralized storage** for quick access with **blockchain-based decentralized validation** for data integrity.
- Implemented **role-based access control** allowing doctors, patients, and hospitals to securely create and share records.
- Integrated **hashing and transaction validation** to prevent unauthorized manipulation and ensure traceability.
- Led the **system design and testing** phases, strengthening skills in **blockchain architecture**, **data security**, and **hybrid system management**.

PUBLICATION

A Review on Machine Learning-Based Algorithms for Heart Disease Diagnosis and Prediction – 2022

• Reviewed machine-learning algorithms for heart-disease diagnosis and prediction, including KNN, Decision Trees, Logistic Regression, SVM, Random Forest, and Naive Bayes.

DOI:10.32628/CSEIT228686

LEARDERSHIP SKILLS

Media Coordinator – Tech Fest Prayan '22 | April 2022

- Led the student media and promotion team for the annual college tech fest.
- Developed creative campaign ideas, poster designs, and visual concepts to attract participation.
- Coordinated between design, content, and logistics teams to ensure consistent event branding and outreach.

Sponsorship Coordinator – Tech Fest Prayan '23 | April 2023

- Managed the student sponsorship team and maintained professional relationships with event sponsors.
- Conducted meetings and negotiations to secure sponsorships and fulfil sponsor requirements.
- Promoted sponsor brands effectively through coordinated media campaigns and on-site representation.

LANGUAGES

English – C1 Level (Professional Proficiency)
German – B1 Level (Intermediate Proficiency)